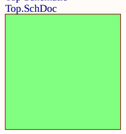
# SOW2 - Rear and Trailer Camera

Top Schematic
Top.SchDoc



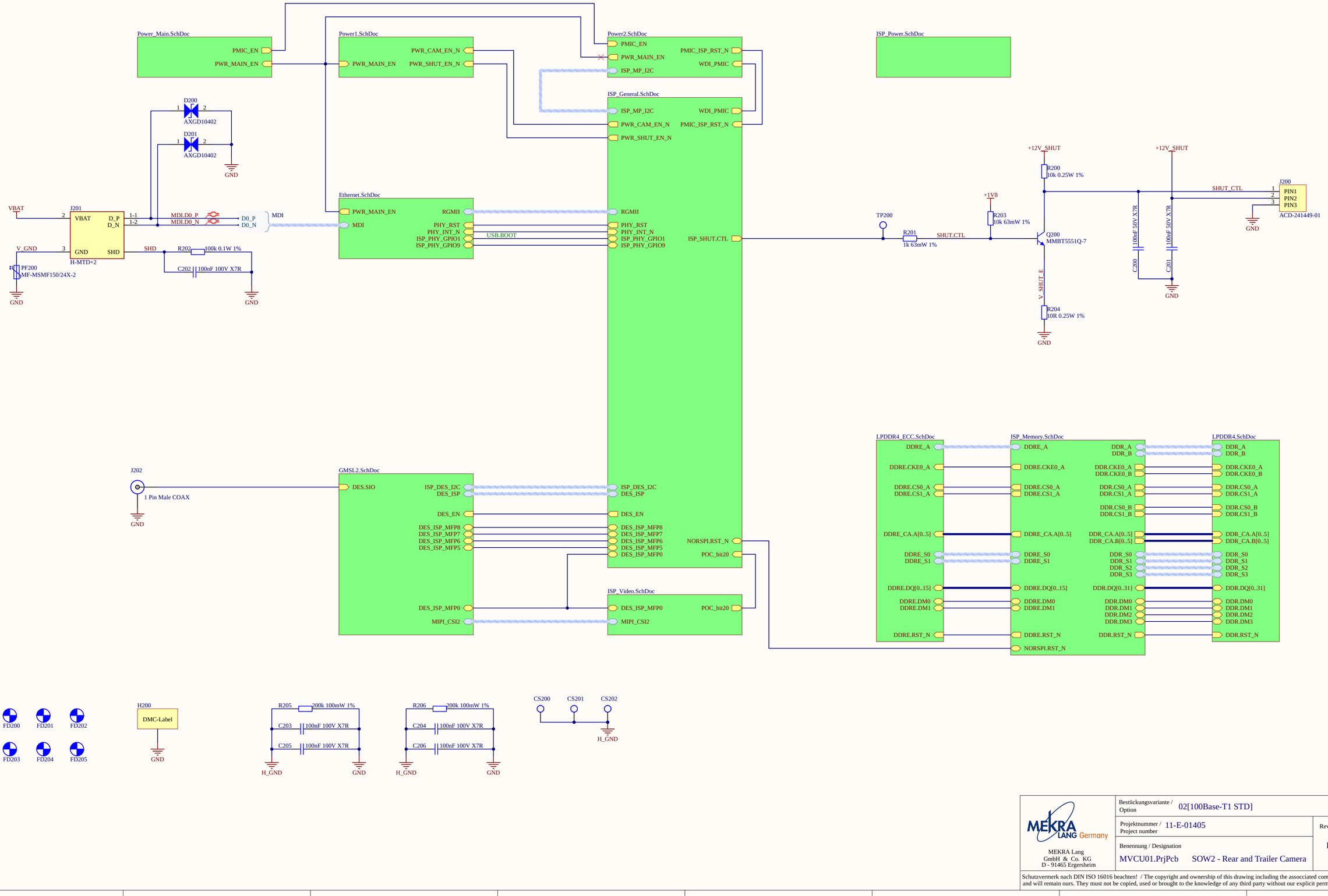
| REV. | DATE | NAME | SCHEET NO. | TITLE / DESCRIPTION OF CHANGE | REASON FOR CHANGE |
|---|---|---|---|---|---|
| A | 15.11.2024 | A. Dünisch, A. Sheqiri | - | Initial Revision | - |
| B | 21.11.2024 | A. Sheqiri | 10-13 | - Changed ManufacturerPartNo of U1000 from CV22FS55 to CV22FS85 | - |
| | | | 6 | - Removed the parameter Clearance - Constraint GAP=0.4mil of signal DES.SIO | |
| C | 02.12.2024 | A. Sheqiri | 12 | - Changed R1230 to 26.1R | - |
| D | 01.05.2025 | A. Dünisch | 9, 12 | - Removed A-Sample debug parts | - Redesign |
| | | | 6 | - Added Testpoints TP603-606 to GMSL2 power supply | |
| | | | 2 | - Changed J200 to ACD-241449-01 | |
| | | | 2 | - Changed R200 to 0805 size | |
| | | | 4 | - Changed C418 and C420 to 50V and 0805 size | |
| | | | 9, 10 | - Added U900 and periphery for ECC | |
| | | | 6, 11, 12 | - Renamed GPIO traces between DES and ISP | |
| | | | 4 | - Changed L400 to 5.6uH | |
| | | | 3 | - Changed L301 to 15uH | |
| | | | 5 | - Swapped +0V8 and +3V3 | |
| | | | 5 | - Changed U501 to TLV73333PQDBVRQ1 | |
| | | | 5 | - Added U502 (TLV73318PQDBVRQ1), C538-541, R522-524 | |
| | | | 7, 12 | - Added PHY_INT_N and ISP_PHY_GPIO1 nets | |
| | | | 12 | - Removed pin header for I2C | |
| | | | 12 | - Replaced Y1201 by ECS-.327-12.5-34QCS-TR | |
| | | | 12 | - Added Y1202 as fitting option for RTC, FB1201, C1218, R1267-1269 | |
| E | 07.05.2025 | A. Dünisch | 4 | - Updated PartNo of U400 to MPQ4232AGVE-0015-AEC1-Z | - MEKRA custom number 0015 |
| | | | 5 | - Updated PartNo of U500 to MPQ70165FSGUTE-0021-AEC1-Z | - MEKRA custom number 0021 |
| | | | 12 | - Swapped pins DES_ISP_MFP5 & 6 with ISP_MP_I2C.CLK & DAT on Ambarella side | - ISP_MP_I2C.CLK & DAT were connected to I2C slave port |
| F | 02.06.2025 | A. Dünisch | 12 | - Added Y1203 (MC2016Z125.0000C19XSH) for external RGMII CLK, C1219, C1220, C1221, FB1202, R1270, R1271 | - Needed for RGMII |
| | | | 12 | - Changed HW-ID to 0b00001 (fitting option) | - B-Sample ID |
| | | | 7 | - Changed fitting options for PHY Pin Strapping | - Internal delay for TXC and RXC needed (length tuning on PCB) |
| | | | 7 | - Added series resistors R720-731 to RGMII | - Recommended by PHY manufactuerer and MAC manufacturer |
| | | | 5 | - Added TP509 to +1V8 | - Possibly needed for flash programming in production |
| | | | 3 | - Added R321 and net MAIN.RST_N to U301 pin 2 | - Regulator U301 switches off when overvoltage on VBAT |
| G | 27.06.2025 | A. Dünisch | 10 | - Connected NORSPI.RST_N to pin AA6 of U1000 via 0R R1002 | - Recommended by Ambarella |
| | | | 12 | - Connected NORSPI.RST_N to pin Y20 of U1000 via 0R R1273 (not fitted) | - Recommended by Ambarella |
| | | | 12 | - Added pull-up R1272 to net USB.DETECT | - Recommended by Ambarella |
| | | | 12 | - Moved RGMII nets from ENET1 port to ENET port on U1000 (B12, E15, E14, D14, E16, D13, C11, E13, E12, C14, A10, C12, C13, A11, D12 ---> A13, K19, G17, H18, F17, E17, L19, H17, K17, J17, K18, L18, J18, B11, L17) | - Recommended by Solectrix and Ambarella |
| | | | 12 | - Moved nets ISP_SHUTTER.CTL, DES_ISP_MFP7, POWER_CAM_EN, POWER_SHUT_EN to other free GPIOs of U1000 (K19, L19, K18, L18 ---> C14, C11, C12, C13) | - Due to change of ENET ports |
| | | | 12 | - Added nets POC_bit4, POC_bit5 and POC_bit6 to pins E14, D14 and E16 of U1000 | - Dedicated Power On Configuration pins which were covered by RGMII pins/nets |
| | | | 12 | - POC bits 21, 0, 1, 2, 3 (A13, G17, H18, F17, E17) are now covered by RGMII pins/nets | - Dedicated Power On Configuration pins which are now covered by RGMII pins/nets |
| | | | 13 | - Added C1357-59 to +0V75 net | - Due to layout-changes |
| H | 07.07.2025 | A. Dünisch | 8 | - Changed C824-833 and C842-847 from 100nF 6.3V X7R to 1uF 4V X7R size 0402 each | - Recommended by Ambarella |
| | | | 9 | - Changed C924-933, C941-946 and C948 from 100nF 6.3V X7R to 1uF 4V X7R size 0402 each | - Recommended by Ambarella |
| | | | 13 | - Changed C1305-1307, C1309-1310, C1314-1315, C1319-1320, C1343-1344, C1346-1350, C1352-1353 and C1357 from 100nF 6.3V X7R to 1uF 4V X7R size 0402 each | - Recommended by Ambarella |
| I | 01.12.2025 | A. Dünisch | 2, 12 | - Renamed SHUTTER_CTL nets to SHUT_CTL | - Shorter and uniform naming |
| | | | 2 | - Added contact springs CS200-CS202 (not fitted) | - For contact to heat spreader in housing |
| | | | 3 | - Replaced D300, D301 and D303 with BAT54C; added D310 (BAT54C) to nets REG12V_BST and VPOL_G_BST | - Performance reasons |
| | | | 3 | - Replaced Q300 and Q301 with Dual MOSFET SQJ980AEP-T1_GE3 | - Performance reasons |
| | | | 4 | - Added R426 to nets REG12V_BST and REG12V_BST1 | - Fitting option |
| | | | 4 | - Change R407 to 90.9k and R408 to 10k | - Corrected resistor devider |
| | | | 4, 12 | - Renamed nets POWER_CAM_EN and POWER_SHUT_EN to PWR_CAM_EN_N and PWR_SHUT_EN_N; added TP408 and TP409 to these nets | - Because the signal gets inverted |
| | | | 5 | - Added TP514 to net WDI_PMIC | - For testing reasons |
| | | | 7 | - Changed C700, C709-10 to 10V voltage rating | - Performance reasons |
| | | | 7 | - Added R734 to nets PHY.INH and PWR_MAIN_EN | - For current limitation |
| | | | 7 | - Removed net ISP_PHY_GPIO0; added net ISP_PHY_GPIO9 | - Problems with GPIO0 standard configuration |
| | | | 7 | - Added R732 to net ISP_PHY_GPIO9 and R733 to net ISP_PHY_GPIO1 | - Fitting options for GPIOs |
| | | | 7 | - Added TP713 and TP714 to SMI and TP715 to PHY.GPIO2 and TP716 to PHY.GPIO0 | - For testing reasons |
| | | | 7 | - Added S700 (not fitted), D700-D702 and R735 | - For testing and development reasons |
| | | | 7, 12 | - Renamed net PHY_RST_N to PHY_RST | - Because the signal gets inverted |
| | | | 8 | - Changed C801, C812-15, C825-28 to 4.7uF 6.3V X7R (0603) | - Performance reasons |
| | | | 8 | - Changed C810 and C823 to 10uF 6.3V X7R (0805) | - Performance reasons |
| | | | 8 | - Changed C821, C835-40 to 1uF 4V X7R (0402) | - Performance reasons |
| | | | 8 | - Changed C824, C829 and C830 to 100nF 6.3V X7R (0402) | - Performance reasons |
| | | | 8 | - Added C841 (100nF 6.3V X7R 0402) and C848-55 (1uF 4V X7R 0402) | - Performance reasons |
| | | | 9 | - Changed C901, C911-14, C924-27 to 4.7uF 6.3V X7R (0603) | - Performance reasons |
| | | | 9 | - Changed C910 and C923 to 10uF 6.3V X7R (0805) | - Performance reasons |
| | | | 9 | - Changed C920-21, C935-40 and C947 to 1uF 4V X7R (0402) | - Performance reasons |
| | | | 9 | - Changed C928-32 to 100nF 6.3V X7R (0402) | - Performance reasons |
| | | | 9 | - Added C949-56 (1uF 4V X7R 0402) | - Performance reasons |
| | | | 11 | - Renamed net DES_ISP.MFP0 to POC_bit19 | - Shorter and uniform naming |
| | | | 11 | - Changed R1101 to 0R (0603); added R1102 (10k 0402) to nets POC_bit19 and GND | - For testing and development reasons |
| | | | 12 | - Removed R1253 from DES_ISP_MFP0 net | - Replaced by R1102 in principle |
| | | | 12 | - Removed ISP_PHY_GPIO0 and connected ISP_PHY_GPIO1 net to pin L17 (POC_bit10/USB BOOT); added TP1233 | - Problems with GPIO0 standard configuration |
| | | | 12 | - Connected ISP_PHY_GPIO9 net to pin D10 (former PHY_GPIO1) | - Replaces "old" GPIO1 |
| | | | 12 | - Connected DES_ISP_MFP0 to ENET1_PTP_PPS net via R1275 (0R 0603) and added TP1236 | - For testing and development reasons |
| | | | 12 | - Added TP1234 to DES_ISP_MFP7, TP1232 to DES_ISP_MFP8, TP1231 to DES_ISP_MFP6 and TP1235 to DES_EN | - For testing and development reasons |
| | | | 13 | - Changed C1302 to 10uF 6.3V X7R (0805) | - Performance reasons |
| | | | 13 | - Changed C1305-7, C1309-10, C1343-44, C1346-47, C1357-59 to 4.7uF 2.5V X7T (0402) | - Performance reasons |
| | | | 13 | - Changed C1303-04 to 10uF 2.5V X7T (0603) | - Performance reasons |
| | | | 13 | - Added C1360 (4.7uF 2.5V X7T 0402) and C1361 (100nF 6.3V X7R 0402) | - Performance reasons |
| | | | 13 | - Changed C1312, C1314-15, C1317, C1319-20 to 4.7uF 6.3V X7R (0603) | - Performance reasons |
| | | | 13 | - Changed C1348-50, C1352-53 to 2.2uF 64V X7S (0402) | - Performance reasons |
| J | 22.05.2026 | A. Schwinger | 2 | - Switched signals MDLD0_P and MDLD0_N at J201 -> 1-1 = D_P, 1-2 = D_N | - Specified by Customer |
| | | | 2 | - Add contact pads to H_GND (Compelma SSMD011032007) | - High-impedance connection between the heat spreader inside the housing and the PCB |
| | | | 3 | - Change D302 to SMBJ60CA-Q (Bourns) | - Specifying a particular component (previously generic) |
| | | | 3 | - Change the 5V regulator output to 4.5V -> Adjust feedback resistor R2 (R314) to 8.87 kΩ | - Compliance with the PMIC's input range, taking into account the regulator's output tolerance |
| | | | 3 | - Add C326 (10uF 50V X7R 1210) to VBAT_FIL - not fitted | - Option to retrofit if needed |
| | | | 3 | - Add Zenerdiode (BZT52C5V6Q-7-F) + Resistor (10R) in parallel to 4.5V reg. output | - FuSa, prevent the 4.5V regulator output from exceeding 5.6V |
| | | | 3 | - L300 -> Taiyo Yuden LCDND4040WKT100MM | - Performance reasons |
| | | | 3 | - FB300 -> TDK MPZ2012S102ATD25 | - Performance reasons |
| | | | 4 | - Schottky diodes on highside switch outputs (NRVB0540T1G) | - Protecting the switch (shutter and camera power supply) from short-circuiting the connector pins to VBAT |
| | | | 5 | - FB505 -> BLM18SP300SH1D (30 ohms) | - Performance reasons |
| | | | 5 | - Change the designation of all ferrites from 33 ohms to 30 ohms | - Naming error |
| | | | 5 | - Remove 1V8_AO LDO | - Was an assembly option for V_ISP_RTC_VDD; never used |
| | | | 5 | - Add 3V0 LDO for V_ISP_AVDD33 and V_ISP_3V0_VDDO | - The circuits were previously powered by 3V3 -> 3V3 is the maximum voltage according to the CV22FS datasheet |
| | | | 5 | - Voltage monitoring added for PMIC outputs (MPS MPQ79501-AEC1) | - FuSa |
| | | | 5 | - Ext. temperature sensor added (TI TMP390AQDRLRQ1) | - FuSa |
| | | | 5 | - Swap the PMIC outputs<br>1V1: Buck3 -> Buck1<br>1V2: Buck6 -> Buck2<br>0V75: Buck1+2 -> Buck 3+4<br>0V8: Buck4 -> Buck6<br>1V8: Buck5 | - The 0V75 output must drive the highest current -> Buck3 and Buck4 can drive the highest current (4A each) -> better performance, recommended by the manufacturer MPS |
| | | | 6 | - L600 -> Murata LQW18CNR21J0HD<br>- L601 -> Murata LQW32FT2R2M0HL<br>- L602 -> Murata LQH3NPH150MMEL | - Previously selected Murata inductors were specifically designed for PoC, but only for automotive infotainment/comfort equipment. Change to automotive powertrain/safety equipment also specifically for PoC. Change made in accordance with specifications from Analog Devices |
| | | | 12 | - Pull-Down resistor (100k) added to net USB.DETECT | - USB is only used in development |
| | | | 12 | - Remove crystal for RTC | - Was an assembly option for RTC; never used |
| | | | 13 | - Connected V_ISP_MIPI_AVDD/MIPI_AVDD18_IO to 1V2 | - Previously set to 1V1 -> too close to the lower limit of the voltage range (1.07V - 1.3V) |
| K | 08.06.2026 | A. Schwinger | 3 | - Change C320 to 50V rated voltage and 0805 package | - Previous voltage rating too low |
| | | | 12 | - Change U1201 to TI SN74LVC1G79QDCKRQ1 | - Previous part TI SN74LVC1G175DCKR was not AEC-Q |
| | | | 3-12 | - Change all 100nF 0402 Capacitors to 50V voltage rating | - Reduction of variance |

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| Projektnummer / Project number | 11-E-01405 | Revision | Blatt / sheet |
| Benennung / Designation | MVCU01.PrjPcb SOW2 - Rear and Trailer Camera | K | 1 of 13<br>Rev. History.SchDoc |

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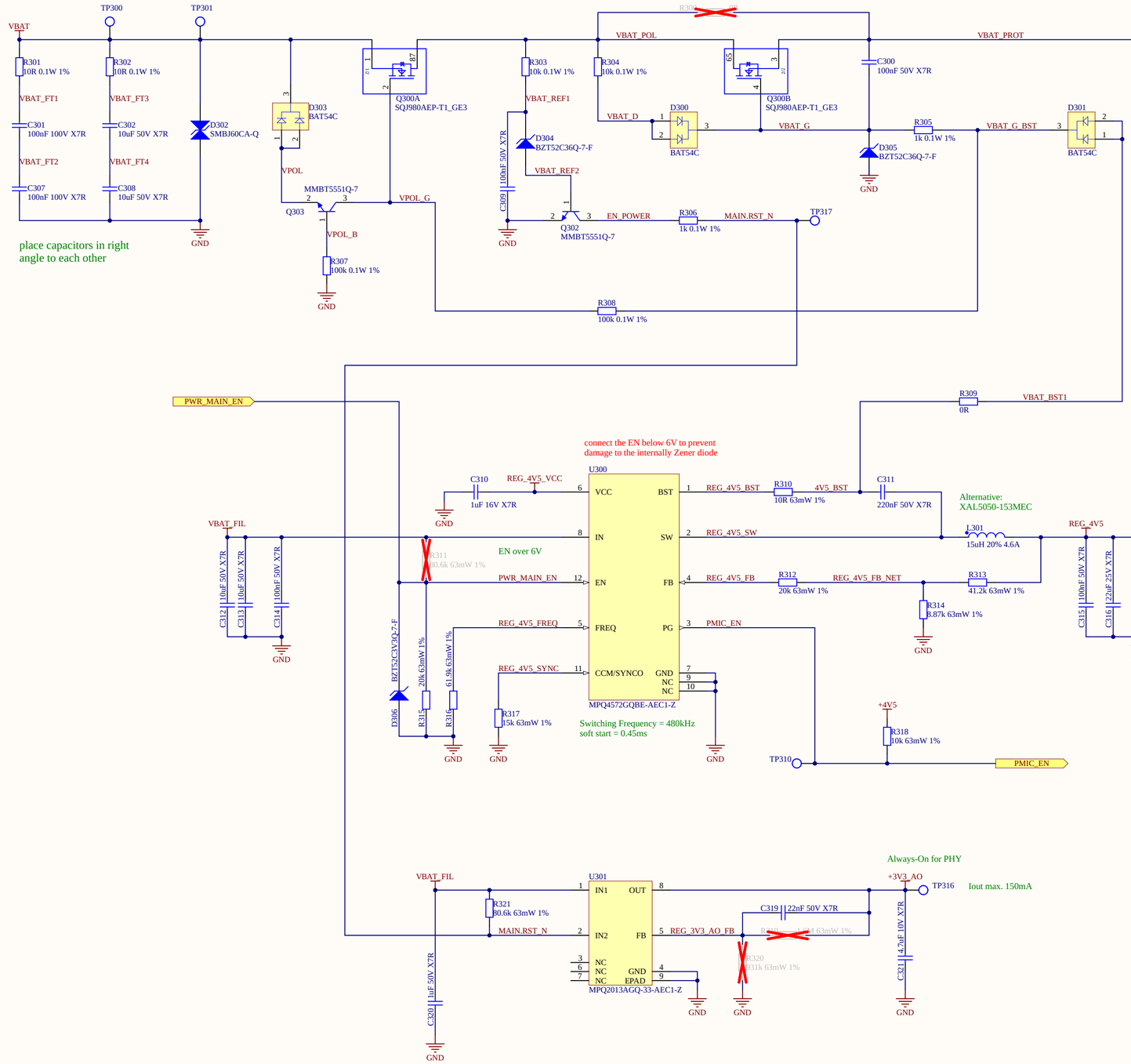
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| | Projektnummer / Project number: 11-E-01405 | Revision: K | Blatt / sheet: 2 of 13 |
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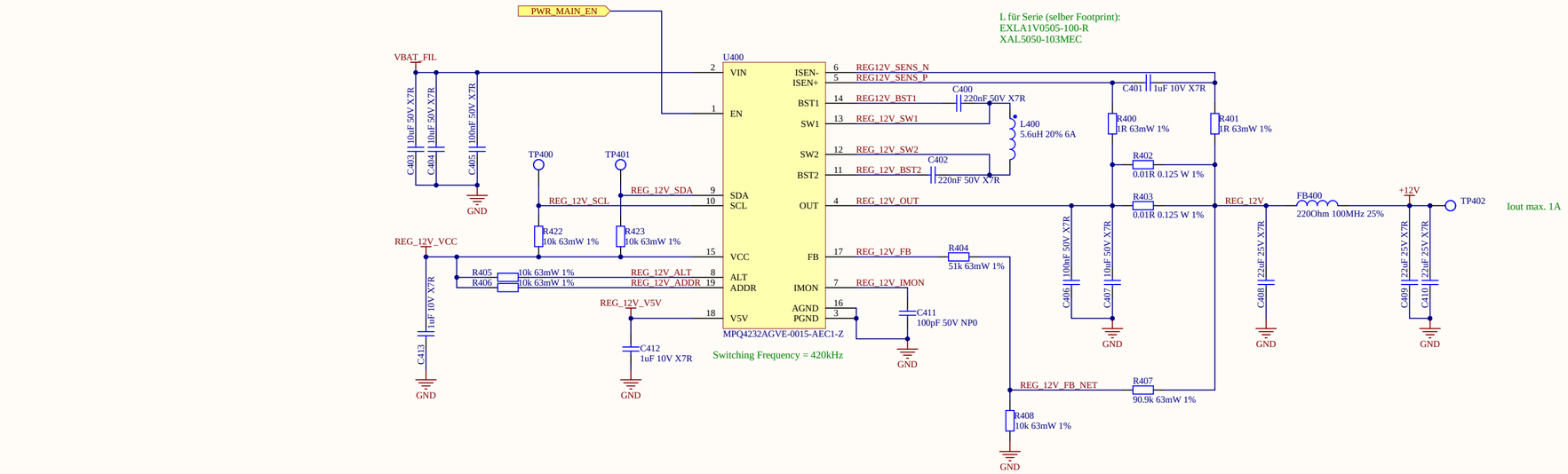
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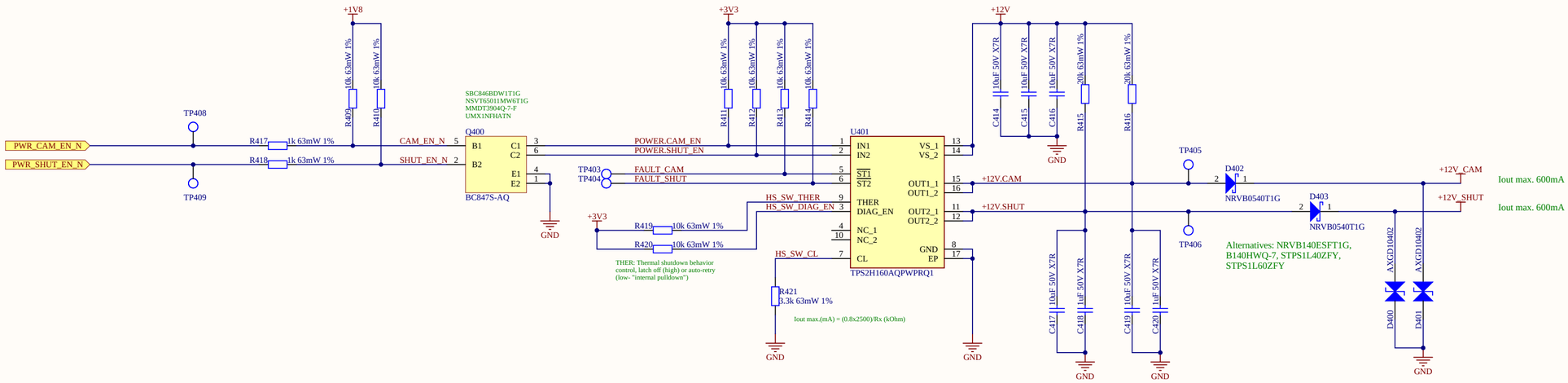



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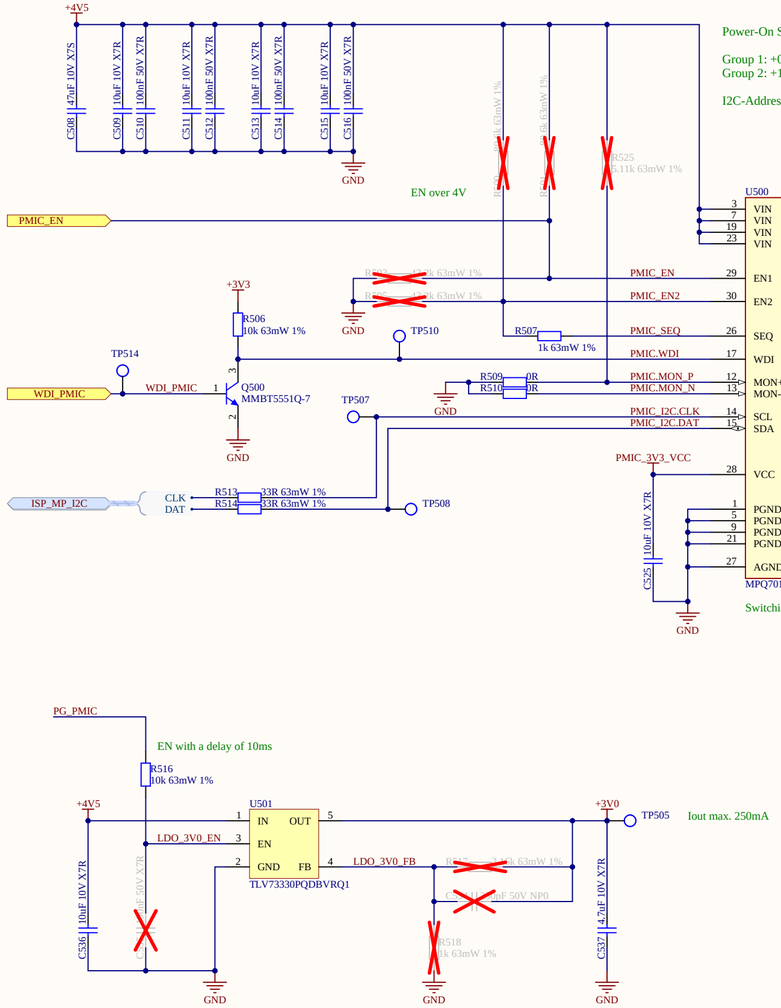


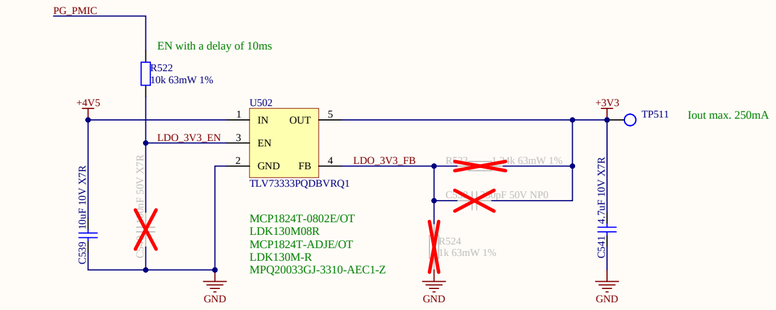




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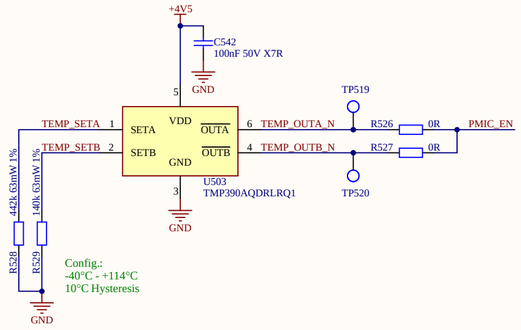




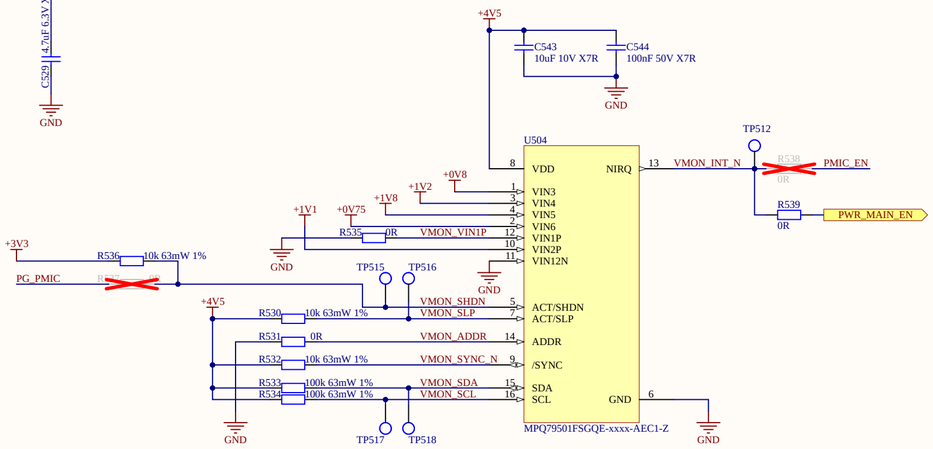


## Temperature Sensor




## Voltage Monitor

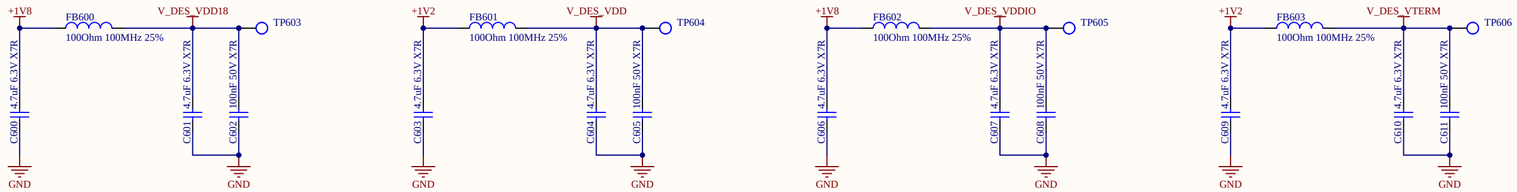
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| | Projektnummer / Project number: 11-E-01405 | Revision: K | Blatt / sheet: 5 of 13 |
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## MIPI_PHY3 (0x333)

MIPI Lane Swapping

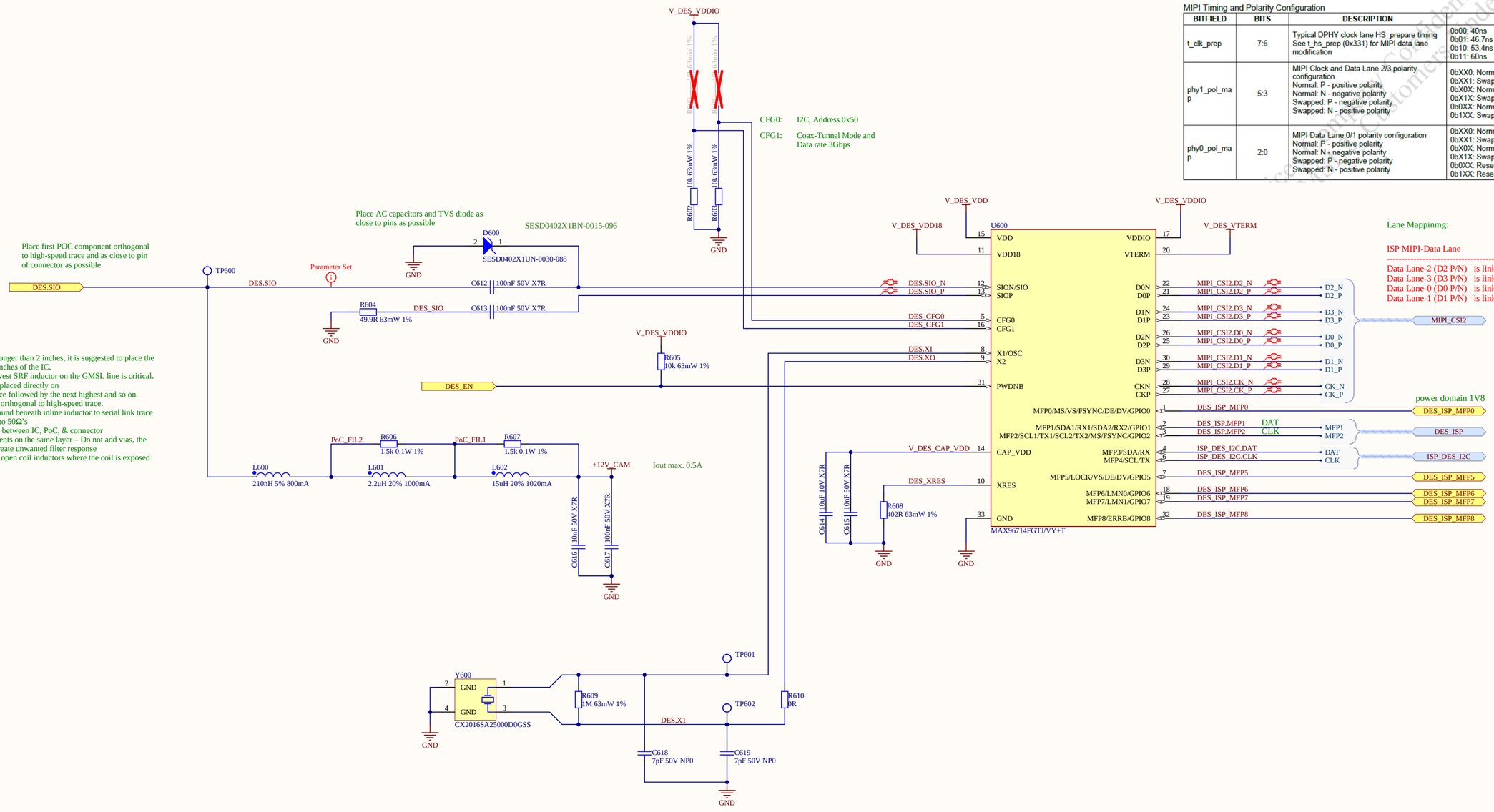
| BITFIELD | BITS | DESCRIPTION | DECODE |
|---|---|---|---|
| phy1_lane_map | 7:4 | Lane mapping for Data Lane 2 and 3 | 0bXX00: D2P/N is assigned to Data Lane 0<br>0bXX01: D2P/N is assigned to Data Lane 1<br>0bXX10: D2P/N is assigned to Data Lane 2 (default)<br>0bXX11: D2P/N is assigned to Data Lane 3<br>0b00XX: D3P/N is assigned to Data Lane 0<br>0b01XX: D3P/N is assigned to Data Lane 1<br>0b10XX: D3P/N is assigned to Data Lane 2<br>0b11XX: D3P/N is assigned to Data Lane 3 (default) |
| phy0_lane_map | 3:0 | Lane mapping for Data Lane 0 and 1 | 0bXX00: D0P/N is assigned to Data Lane 0 (default)<br>0bXX01: D0P/N is assigned to Data Lane 1<br>0bXX10: D0P/N is assigned to Data Lane 2<br>0bXX11: D3P/N is assigned to Data Lane 3<br>0b00XX: D1P/N is assigned to Data Lane 0<br>0b01XX: D1P/N is assigned to Data Lane 1 (default)<br>0b10XX: D1P/N is assigned to Data Lane 2<br>0b11XX: D1P/N is assigned to Data Lane 3 |

## MIPI_PHY5 (0x335)

MIPI Timing and Polarity Configuration

| BITFIELD | BITS | DESCRIPTION | DECODE |
|---|---|---|---|
| t_clk_prep | 7:6 | Typical DPHY clock lane HS_prepare timing<br>See t_hs_prep (0x331) for MIPI data lane modification | 0b00: 40ns<br>0b01: 46.7ns<br>0b10: 53.4ns<br>0b11: 60ns |
| phy1_pol_map | 5:3 | MIPI Clock and Data Lane 2/3 polarity configuration<br>Normal: P - positive polarity<br>Normal: N - negative polarity<br>Swapped: P - negative polarity<br>Swapped: N - positive polarity | 0bXX0: Normal polarity D2P/N<br>0bXX1: Swapped polarity D2N/P<br>0bX0X: Normal polarity D3P/N<br>0bX1X: Swapped polarity D3N/P<br>0b0XX: Normal polarity CKP/N<br>0b1XX: Swapped polarity CKN/P |
| phy0_pol_map | 2:0 | MIPI Data Lane 0/1 polarity configuration<br>Normal: P - positive polarity<br>Normal: N - negative polarity<br>Swapped: P - negative polarity<br>Swapped: N - positive polarity | 0bXX0: Normal polarity D0P/N<br>0bXX1: Swapped polarity D0N/P<br>0bX0X: Normal polarity D1P/N<br>0bX1X: Swapped polarity D1N/P<br>0b0XX: Reserved<br>0b1XX: Reserved |

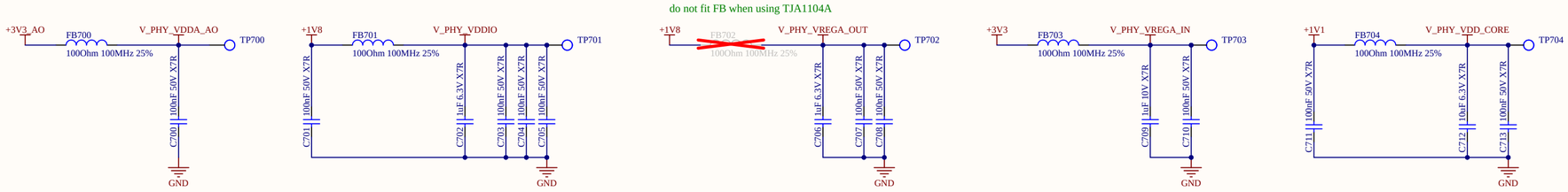
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| | Projektnummer / Project number: 11-E-01405 | Revision: K | Blatt / sheet: 6 of 13 |
| | Benennung / Designation: MVCU01.PrjPcb SOW2 - Rear and Trailer Camera | | GMSL2.SchDoc |

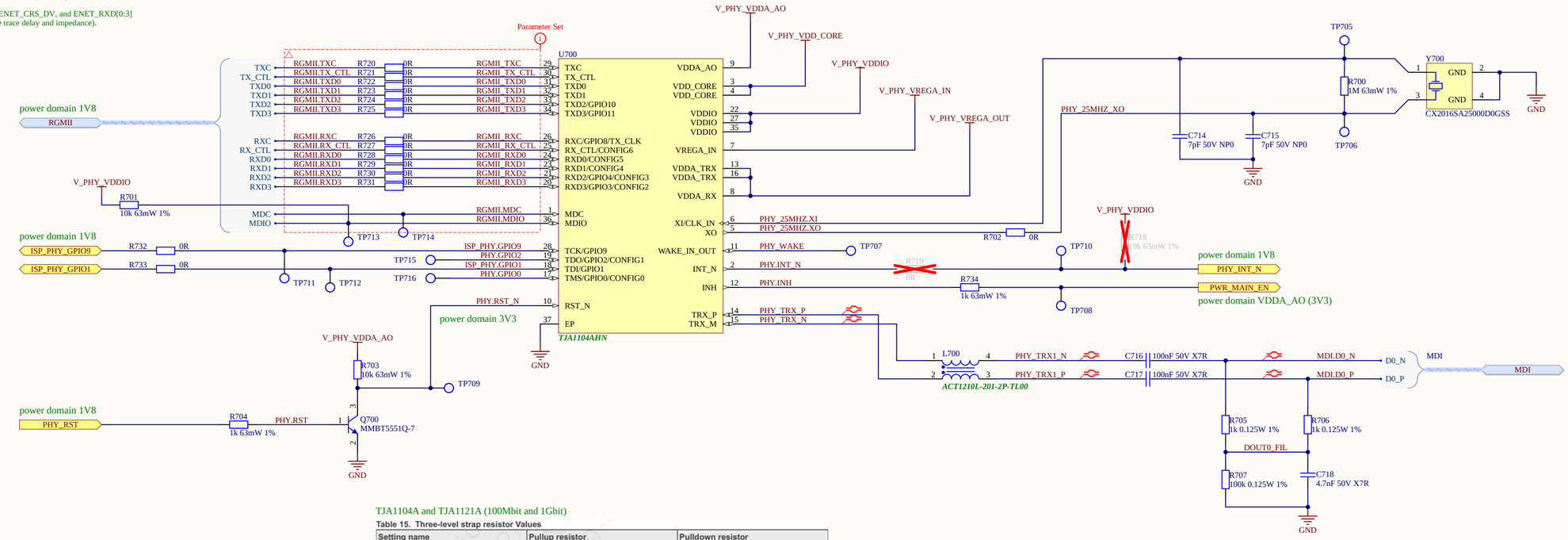
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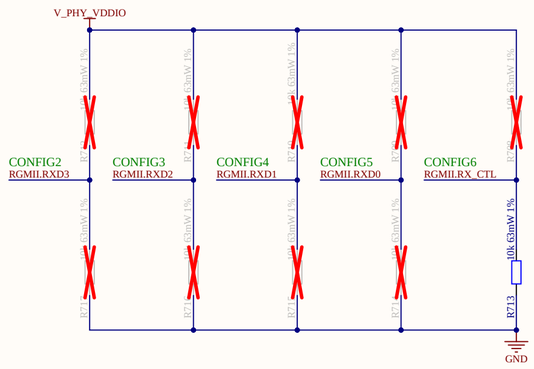
A2




**Ethernet**

- Transmit signal list: Ethernet pins ENET_TXD_[0:3] and ENET_TXEN
  - Keep all signal traces to the same length (or same trace delay and impedance).
  - Length matching is to +/- 100 mils (2.54mm).
- Receive signal list: Ethernet pins ENET_RX_ER, ENET_CRS_DV, and ENET_RXD[0:3]
  - Keep all signal traces to the same length (or same trace delay and impedance).
  - Length matching is to +/- 100 mils (2.54mm).




TJA1104A and TJA1121A (100Mbit and 1Gbit)

**Table 15. Three-level strap resistor Values**

| Setting name<br>Abbreviation | Pullup resistor | Pulldown resistor |
|---|---|---|
| Open (O) | Open | Open |
| High (H) | 10 kΩ | Open |
| Low (L) | Open | 10 kΩ |

**Table 19. PHY address pin-strapping**

| PHY address | Pin CONFIG0 | Pin CONFIG1 | Pin CONFIG2 |
|---|---|---|---|
| 1 | H | H | H |
| 2 | H | H | L |
| 3 | H | H | O |
| 4 | H | L | H |
| 5 | H | L | L |
| 6 | H | L | O |
| 7 | H | O | H |
| 8 | H | O | L |
| 9 | H | O | O |
| 10 | L | H | H |
| 11 | L | H | L |
| 12 | L | H | O |
| 13 | L | L | H |
| 14 | L | L | L |
| 15 | L | L | O |
| 16 | L | O | H |
| 17 | L | O | L |
| 18 | L | O | O |
| 19 | O | H | H |
| 20 | O | H | L |
| 21 | O | H | O |
| 22 | O | L | H |
| 23 | O | L | L |
| 24 | O | L | O |
| 25 | O | O | H |
| 26 | O | O | L |
| 27 | O | O | O |

-> only Pin 20 (CONFIG2) has pin strap resistors

TJA1121A (1Gbit)

**Table 14. Pinstrap pin to function mapping**

| Pin number | Name | Function |
|---|---|---|
| 17 | CONFIG0 | PHY address |
| 19 | CONFIG1 | |
| 20 | CONFIG2 | |
| 21 | CONFIG3 | RGMII/SGMII mode |
| 23 | CONFIG4 | |
| 24 | CONFIG5 | Clock mode |
| 25 | CONFIG6 | 1000BASE-T1 Leader/Follower, Autonomous |

**Table 20. xMII pin-strapping**

| RGMII Mode (TJA1121A) | SGMII Mode (TJA1121B) | Pin CONFIG3 | Pin CONFIG4 |
|---|---|---|---|
| RGMII-ID (RXC) | SGMII-PHY | O | O |
| - | SGMII-MAC | O | H |
| - | - | O | L |
| RGMII | | H | O |
| - | - | H | H |
| - | - | H | L |
| RGMII-ID (TXC/RXC) | - | L | O |
| - | - | L | H |
| JTAG | JTAG | L | L |

-> all RGMII signals on PCB are length tuned within 1.27mm

**Table 21. Clock pin-strapping**

| CLK select | Pin CONFIG5 |
|---|---|
| differential | O |
| XO mode | H -> external 25MHz crystal |
| single ended | L |

**Table 22. Misc pin-strapping**

| Leader/Follower | Autonomous Mode | Pin CONFIG6 |
|---|---|---|
| Leader | enabled | O |
| Follower | enabled | H |
| register controlled (default: Follower) | disabled | L |

TJA1104A (100Mbit)

**Table 17. Pinstrap pin to function mapping**

| Pin # | Function |
|---|---|
| 17, 19, 20 | PHY address |
| 21, 23 | xMII mode |
| 24, 25 | Misc (MASTER/SLAVE/AUTONOMOUS) |

**Table 21. Pinstrap MODE group**

| xMII Mode (TJA1104A) | xMII Mode (TJA1104B) | Pin CONFIG3 | Pin CONFIG4 |
|---|---|---|---|
| RGMII-ID (RXC) | SGMII-PHY | O | O |
| RGMII | SGMII-MAC | O | H |
| RGMII-ID (both TXC/RXC) | | O | L |
| RMII | | H | O |
| RMII-clkmode | | H | H |
| RMII-rev | | H | L |
| MII | | L | O |
| MII-rev | | L | H |
| JTAG | JTAG | L | L |

-> all RGMII signals on PCB are length tuned within 1.27mm

**Table 22. Pinstrap MODE group**

| MASTER/SLAVE | AUTONOMOUS Mode | Polarity Correction | Pin CONFIG5 | Pin CONFIG6 |
|---|---|---|---|---|
| MASTER | enabled | enabled | O | O |
| MASTER | disabled | enabled | O | H |
| SLAVE | enabled | enabled | O | L |
| SLAVE | disabled | enabled | H | O |
| MASTER | enabled | disabled | H | H |
| MASTER | disabled | disabled | H | L |
| SLAVE | enabled | disabled | L | O |
| SLAVE | disabled | disabled | L | H |
| *reserved* | *reserved* | *reserved* | L | L |

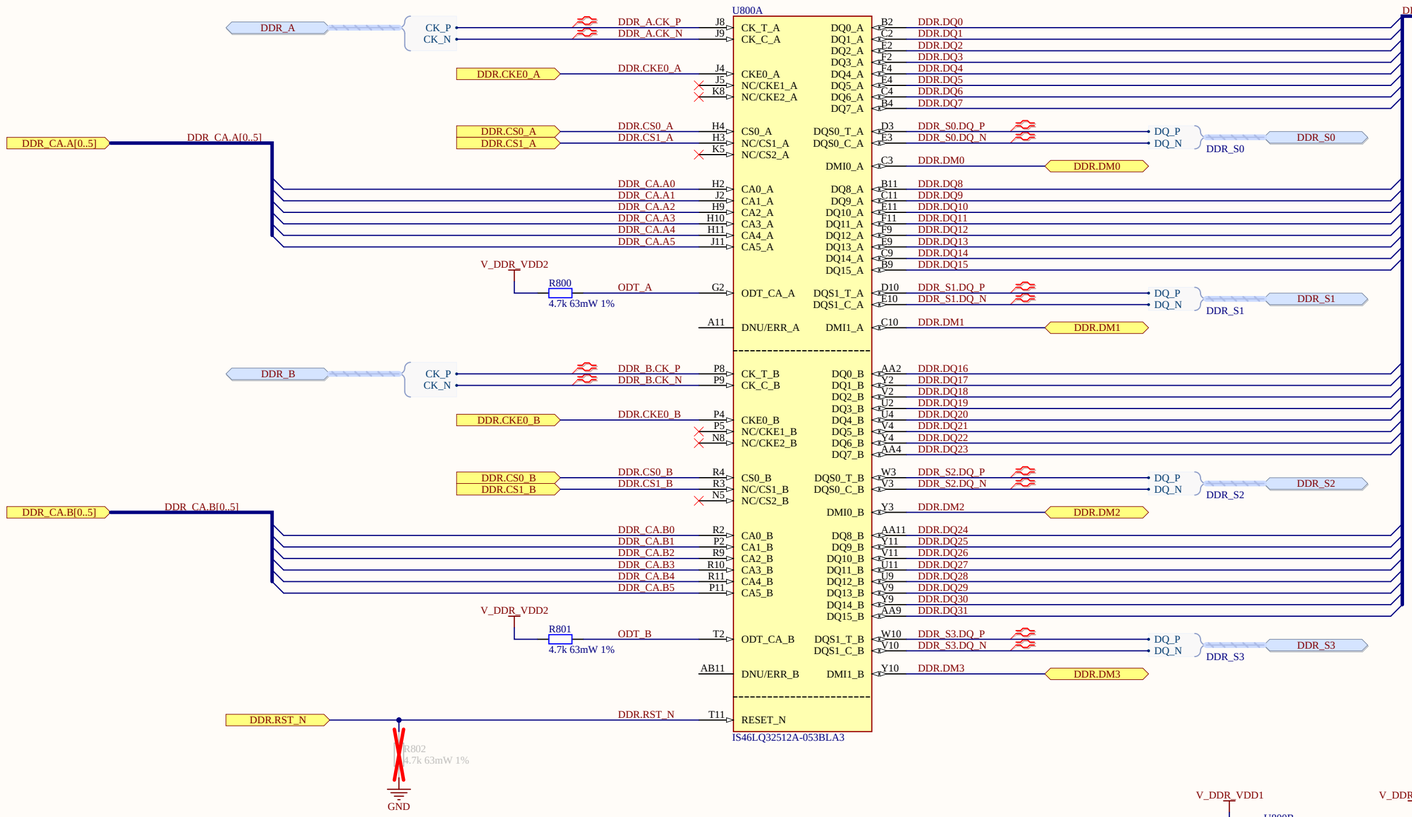
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| | |
|---|---|
| Bestückungsvariante / Option | 02[100Base-T1 STD] |
| Projektnummer / Project number | 11-E-01405 |
| Benennung / Designation | MVCU01.PrjPcb SOW2 - Rear and Trailer Camera |
| Revision | K |
| Blatt / sheet | 7 of 13 |
| | Ethernet.SchDoc |

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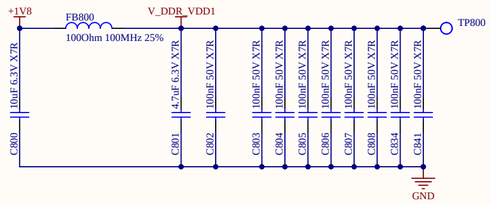
A2

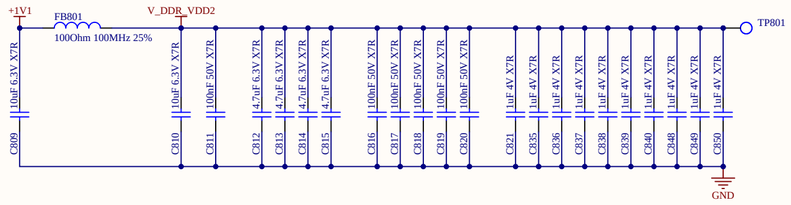


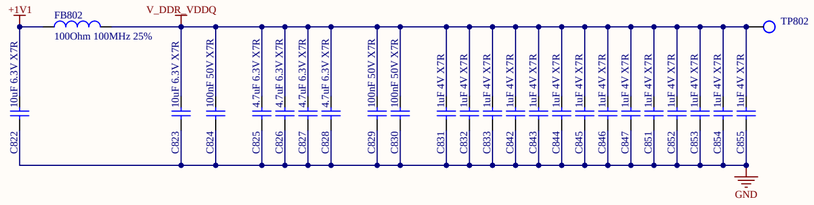

Package delay mit einbinden in Altium (SoC und LPDDR4)

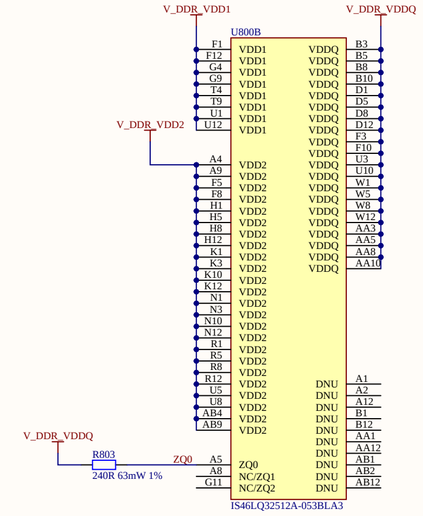
| Signal | Group | PCB + Package Propagation Delay | | Considerations | Convert to Length (Assume Propagation Delay of 160 ps / in) |
|---|---|---|---|---|---|
| | | Minimum | Maximum | | |
| CK_P / CK_N | Clock | Short as possible | 160 ps | [P&N] with +/- 5 mils | < 1000 mils |
| CA<br>CS<br>CKE | Address / Command / Control | CK_P - 16 ps | CK_P + 16 ps | | CK_P +/- 100 mils |
| DQS0_P / DQS0_N | Byte0 DQS | CK_P - 40 ps | CK_P + 40 ps | [P&N] with +/- 5 mils | Within CK_P +/- 250 mils |
| DM0<br>DQ[7:0] | Byte0 Data | DQS0P - 8 ps | DQS0P + 8 ps | | DQ / DM should be in reference to the corresponding DQS within +/- 50 mils |
| DQS1_P / DQS1_N | Byte1 DQS | CK_P - 40 ps | CK_P + 40 ps | [P&N] with +/- 5 mils | Within CK_P +/- 250 mils |
| DM1<br>DQ[15:8] | Byte1 Data | DQS1P - 8 ps | DQS1P + 8 ps | | DQ / DM should be in reference to the corresponding DQS within +/- 50 mils |
| DQS2_P / DQS2_N | Byte2 DQS | CK_P - 40 ps | CK_P + 40 ps | [P&N] with +/- 5 mils | Within CK_P +/- 250 mils |
| DM2<br>DQ[23:16] | Byte2 Data | DQS2P - 8 ps | DQS2P + 8 ps | | DQ / DM should be in reference to the corresponding DQS within +/- 50 mils |
| DQS3_P / DQS3_N | Byte3 DQS | CK_P - 40 ps | CK_P + 40 ps | [P&N] with +/-5 mils | Within CK_P +/- 250 mils |
| DM3<br>DQ[31:24] | Byte3 Data | DQS3P - 8 ps | DQS3P + 8 ps | | DQ / DM should be in reference to the corresponding DQS within +/- 50 mils |

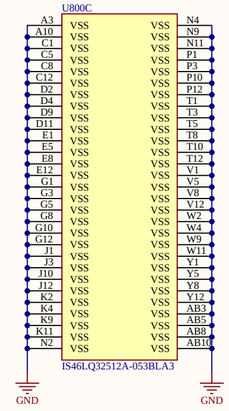
*Table 10-1. LPDDR4 Routing Recommendations.*



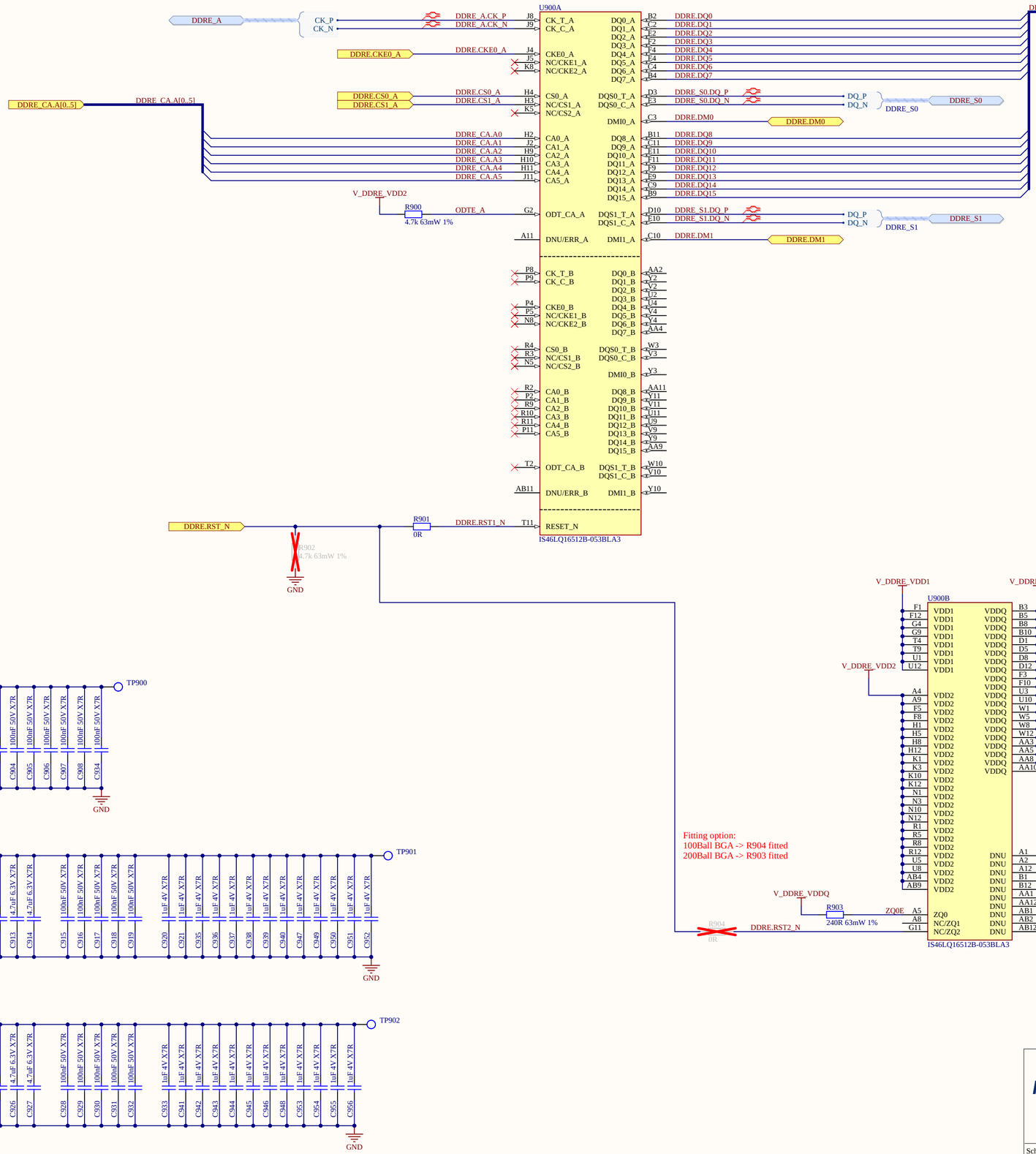




| | | | |
|---|---|---|---|
| MEKRA LANG Germany<br>MEKRA Lang GmbH & Co. KG<br>D - 91465 Ergersheim | Bestückungsvariante / Option: 02[100Base-T1 STD] | | |
| | Projektnummer / Project number: 11-E-01405 | Revision: K | Blatt / sheet: 8 of 13 |
| | Benennung / Designation: MVCU01.PrjPcb SOW2 - Rear and Trailer Camera | | LPDDR4.SchDoc |

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A2




Package delay mit einbinden in Altium (SoC und LPDDR4)

| Signal | Group | PCB + Package Propagation Delay | | Considerations | Convert to Length (Assume Propagation Delay of 160 ps / in) |
|---|---|---|---|---|---|
| | | Minimum | Maximum | | |
| CK_P / CK_N | Clock | Short as possible | 160 ps | [P&N] with +/- 5 mils | < 1000 mils |
| CA<br>CS<br>CKE | Address / Command / Control | CK_P - 16 ps | CK_P + 16 ps | | CK_P +/- 100 mils |
| DQS0_P / DQS0_N | Byte0 DQS | CK_P - 40 ps | CK_P + 40 ps | [P&N] with +/- 5 mils | Within CK_P +/- 250 mils |
| DM0<br>DQ[7:0] | Byte0 Data | DQS0P - 8 ps | DQS0P + 8 ps | | DQ / DM should be in reference to the corresponding DQS within +/- 50 mils |
| DQS1_P / DQS1_N | Byte1 DQS | CK_P - 40 ps | CK_P + 40 ps | [P&N] with +/- 5 mils | Within CK_P +/- 250 mils |
| DM1<br>DQ[15:8] | Byte1 Data | DQS1P - 8 ps | DQS1P + 8 ps | | DQ / DM should be in reference to the corresponding DQS within +/- 50 mils |
| DQS2_P / DQS2_N | Byte2 DQS | CK_P - 40 ps | CK_P + 40 ps | [P&N] with +/- 5 mils | Within CK_P +/- 250 mils |
| DM2<br>DQ[23:16] | Byte2 Data | DQS2P - 8 ps | DQS2P + 8 ps | | DQ / DM should be in reference to the corresponding DQS within +/- 50 mils |
| DQS3_P / DQS3_N | Byte3 DQS | CK_P - 40 ps | CK_P + 40 ps | [P&N] with +/-5 mils | Within CK_P +/- 250 mils |
| DM3<br>DQ[31:24] | Byte3 Data | DQS3P - 8 ps | DQS3P + 8 ps | | DQ / DM should be in reference to the corresponding DQS within +/- 50 mils |

*Table 10-1. LPDDR4 Routing Recommendations.*

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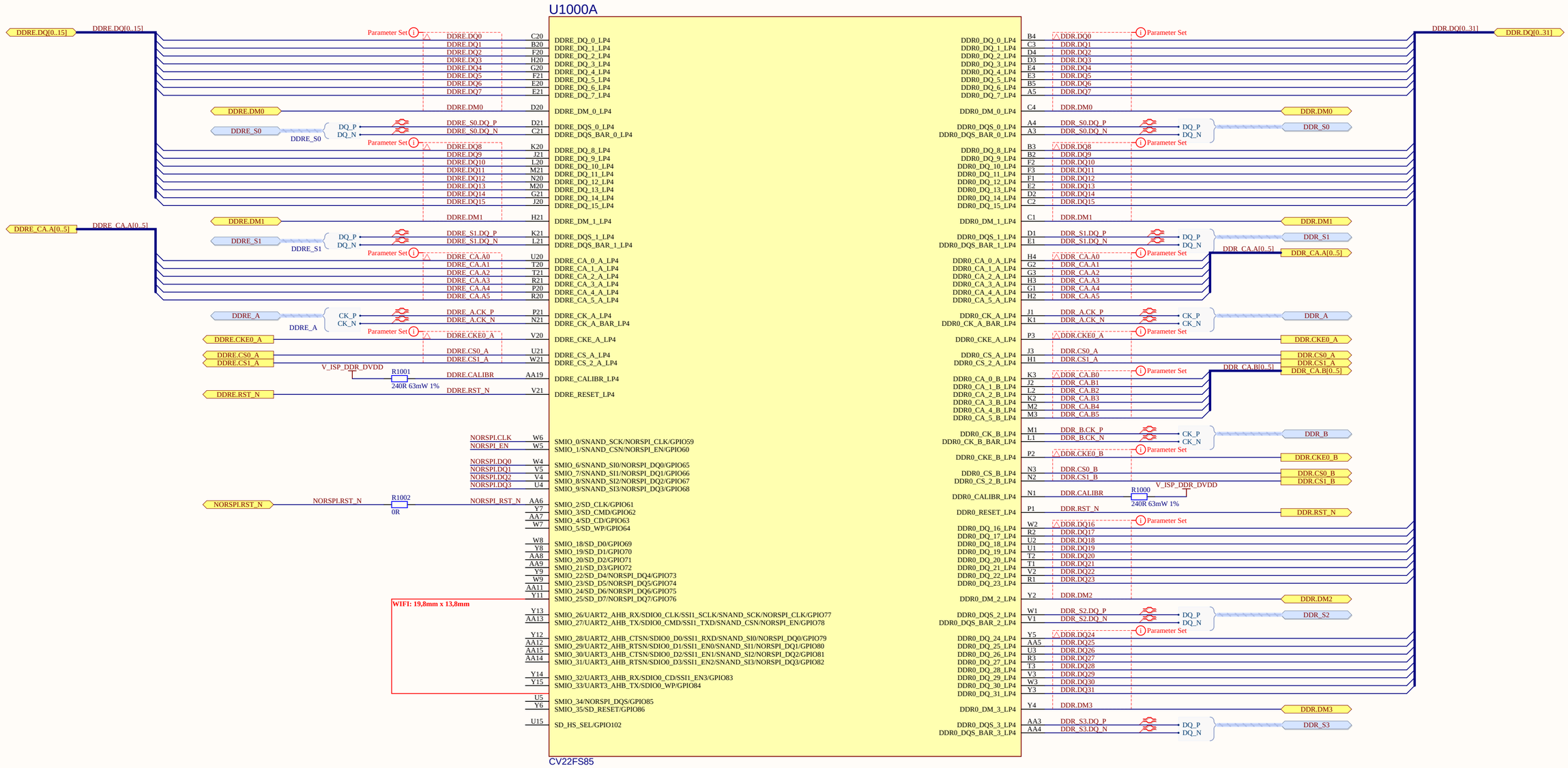
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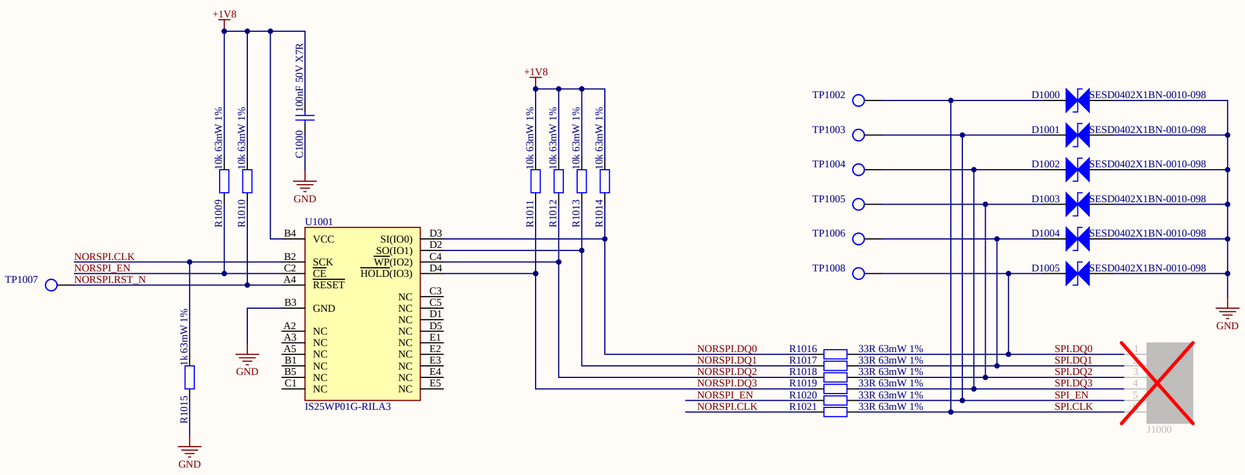
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|---|---|
| Bestückungsvariante / Option | 02[100Base-T1 STD] |
| Projektnummer / Project number | 11-E-01405 |
| Benennung / Designation | MVCU01.PrjPcb SOW2 - Rear and Trailer Camera |
| Revision | K |
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| | LPDDR4_ECC.SchDoc |

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A2




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| Projektnummer / Project number | 11-E-01405 |
| Revision | K |
| Blatt / sheet | 10 of 13 |
| Benennung / Designation | MVCU01.PrjPcb SOW2 - Rear and Trailer Camera |
| | ISP_Memory.SchDoc |

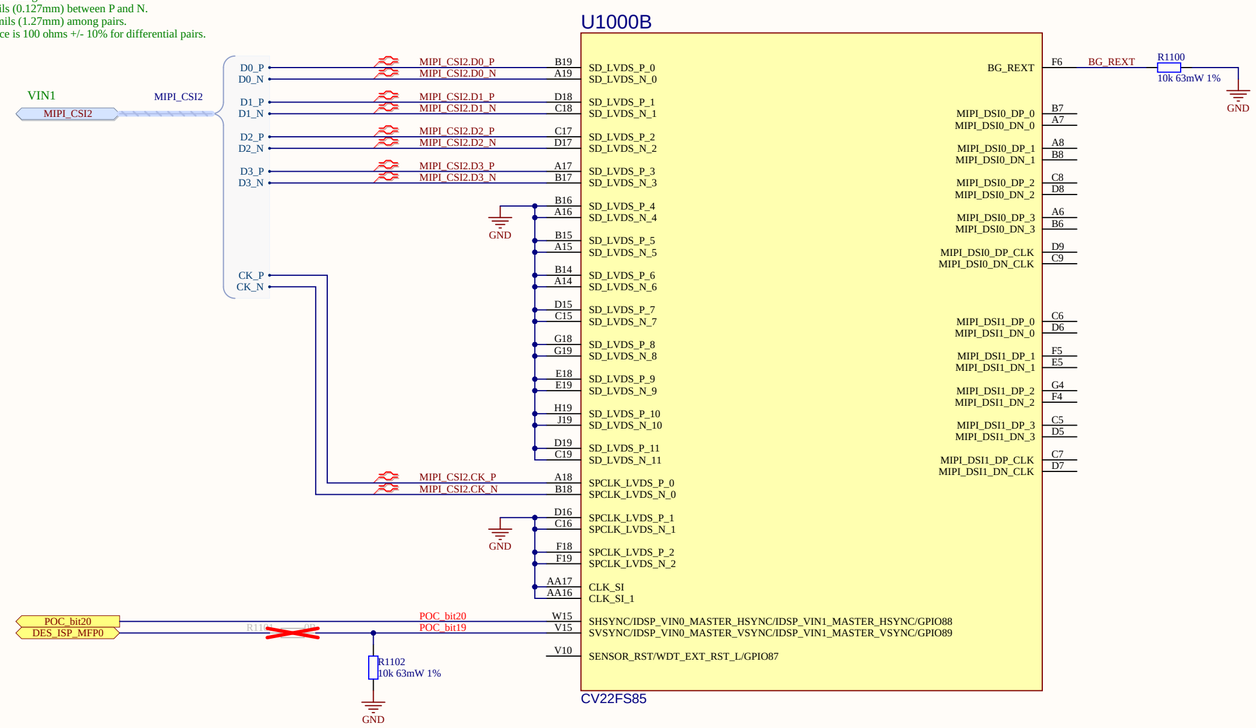
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A2

**VIN: MIPI CSI2**

- Signal list: SD_LVDS_P/N_[0:11], SPCLK_LVDS_P/N_[0:2]
  - Trace lengths are 4500 mils (114.3mm) or less.
  - Length matching is to:
    - +/- 5 mils (0.127mm) between P and N.
    - +/- 50 mils (1.27mm) among pairs.
  - Impedance is 100 ohms +/- 10% for differential pairs.







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## U1000C

CV22FS85

| | | | | |
|---|---|---|---|---|
| USB_DM | | | | |
| USB_DP | | | | |
| DETECT_VBUS | | | | |
| USB_REXT | | | | |
| JTAG_RST_L | | | | |
| JTAG_CLK | | | | |
| JTAG_TMS | | | | |
| JTAG_TDI | | | | |
| JTAG_TDO | | | | |
| UART_APB_RX/EHCI_APP_PRT_OVCURR0/GPIO16 | | | | |
| UART_APB_TX/EHCI_PRT_PWR_0/GPIO17 | | | | |
| UART0_AHB_RX/SSI1_SCLK/EHCI_APP_PRT_OVCURR0/TM11/CLK/PWM10/GPIO90 | | | | |
| UART0_AHB_TX/SSI1_TXD/WDT_EXT_RST_L/GPIO91 | | | | |
| UART0_AHB_CTS_N/SSI1_RXD/EHCI_PRT_PWR0/TM12_CLK/GPIO92 | | | | |
| UART0_AHB_RTS_N/SSI1_EN0/TM13_CLK/GPIO93 | | | | |
| UART1_AHB_RX/TM12_CLK/GPIO94 | | | | |
| UART1_AHB_TX/GPIO95 | | | | |
| UART1_AHB_CTS_N/IDSP_VIN1_MASTER_VSYNC/GPIO96 | | | | |
| UART1_AHB_RTS_N/IDSP_VIN1_MASTER_HSYNC/GPIO97 | | | | |
| XI_RTC | | | | |
| XO_RTC | | | | |
| XIN | | | | |
| XOUT | | | | |
| POR_L | | | | |
| ENET_GTX_CLK/ENET_2ND_REF_CLK/PWM9/GPIO41 | | | | |
| ENET_TXEN/PWM8/GPIO27 | | | | |
| ENET_TXD_0/PWM9/GPIO28 | | | | |
| ENET_TXD_1/PWM10/GPIO29 | | | | |
| ENET_TXD_2/PWM11/GPIO30 | | | | |
| ENET_TXD_3/GPIO31 | | | | |
| ENET_RXD_0/GPIO32 | | | | |
| ENET_RXD_1/GPIO33 | | | | |
| ENET_RXD_2/GPIO34 | | | | |
| ENET_RXD_3/PWM3/GPIO35 | | | | |
| ENET_RXDV/PWM4/GPIO36 | | | | |
| ENET_MDC/PWM5/AHB_MDC/GPIO37 | | | | |
| ENET_MDIO/PWM6/AHB_MDIO/GPIO38 | | | | |
| ENET_CLK_RX/ENET_REF_CLK/PWM8/GPIO40 | | | | |
| ENET_EXT_OSC_CLK/GPIO42 | | | | |
| ENET_2ND_REF_CLK/WDT_EXT_RST_L/SSI0_EN2/GPIO43 | | | | |
| ENET_PTP_PPS/GPIO39 | | | | |
| ENET1_GTX_CLK/ENET_2ND_REF_CLK/EHCI_PRT_PWR_0/CAN5_TX/GPIO58 | | | | |
| ENET1_TXEN/CAN2_RX/GPIO44 | | | | |
| ENET1_TXD_0/SSI5_TXD/PWM11/GPIO45 | | | | |
| ENET1_TXD_1/SSI5_EN0/GPIO46 | | | | |
| ENET1_TXD_2/VSDELAY_HSYNC_OUT_0/GPIO47 | | | | |
| ENET1_TXD_3/VSDELAY_HSYNC_OUT_1/CAN2_TX/GPIO48 | | | | |
| ENET1_RXD_0/SSI5_SCLK/UART2_AHB_RX/SSIS_SCLK/GPIO49 | | | | |
| ENET1_RXD_1/SSI5_RXD/UART2_AHB_TX/SSIS_TXD/GPIO50 | | | | |
| ENET1_RXD_2/IDC5CLK/VSDELAY_VSYNC_OUT_0/UART2_AHB_CTSN/SSIS_TXD/GPIO51 | | | | |
| ENET1_RXD_3/IDC5DATA/VSDELAY_VSYNC_OUT_1/UART2_AHB_RTSN/SSIS_EN/GPIO52 | | | | |
| ENET1_RXDV/EXT_VSYNC_SRC_INPUT/SSI4_SCLK/CAN3_RX/GPIO53 | | | | |
| ENET1_MDC/AHB_MDC1/SSI4_TXD/CAN3_TX/GPIO54 | | | | |
| ENET1_MDIO/AHB_MDIO1/SSI4_RXD/CAN4_RX/GPIO55 | | | | |
| ENET1_CLK_RX/ENET1_REF_CLK/EHCI_APP_PRT_OVCURR0/CAN5_RX/GPIO57 | | | | |
| ENET1_PTP_PPS/EXT_VSYNC_SRC_INPUT/SSI4_EN0/CAN4_TX/GPIO56 | | | | |
| I2S_CLK/SSI1_SCLK/GPIO21 | W19 | ISP_I2S.CLK | TP1205 | |
| I2S_SI_0/SSI1_TXD/GPIO22 | R17 | ISP_I2S.SI | TP1206 | |
| I2S_SO_0/SSI1_EN0/GPIO23 | W18 | POC_bit16 | | |
| SSI1_RXD/EHCI_APP_PRT_OVCURR0/GPIO24 | W17 | ISP_HW_ID_0 | | |
| EHCI_PRT_PWR_0/GPIO25 | R19 | POC_bit17 | | |
| I2S_WS/GPIO26 | V18 | ISP_I2S.WS | TP1210 | |
| CLK_AU | W20 | POC_bit8 | | |
| IDCSCLK/TM12_CLK/SSI0_EN6/VIN_STRIG0/EHCI_APP_PRT_OVCURR0/GPIO1 | V16 | DES_ISP_MFP5 | TP1215 TP1231 | power domain 1V8 |
| IDCSDATA/TM13_CLK/SSI0_EN7/VIN_STRIG1/EHCI_PRT_PWR0/GPIO2 | V17 | DES_ISP_MFP6 | | DES_ISP_MFP5 DES_ISP_MFP6 |
| SSI0CLK/IDC0CLK/PWM4/GPIO3 | U16 | ISP_DES_I2C.CLK | | |
| SSI0MOSI/IDC0DATA/PWM5/GPIO4 | U18 | ISP_DES_I2C.DAT | CLK | ISP_DES_I2C |
| SSI0MISO/IDC1CLK/PWM6/GPIO5 | U17 | DES_ISP.MFP2 | DAT | |
| SSI0EN0/IDC1DATA/PWM7/GPIO6 | V19 | DES_ISP.MFP1 | MFP2 | DES_ISP |
| SSI0EN1/WDT_EXT_RST_L/SDIO0_HS_SEL/GPIO7 | U19 | | MFP1 | |
| SSI2CLK/IDC2CLK/SSI0_EN2/VSDELAY_VSYNC_OUT_0/GPIO8 | N17 | | | |
| SSI2MOSI/IDC2DATA/SSI0_EN3/VSDELAY_VSYNC_OUT_1/GPIO9 | T18 | | | |
| SSI2MISO/IDC3CLK/SSI0_EN4/VSDELAY_HSYNC_OUT_0/GPIO10 | T19 | | | |
| SSI2EN0/IDC3DATA/SSI0_EN5/VSDELAY_HSYNC_OUT_1/GPIO11 | R18 | | | |
| SSI3CLK/IDC4CLK/SSIS_SCLK/UART3_AHB_RX/GPIO12 | P18 | ISP_MP_I2C.CLK | CLK | |
| SSI3MOSI/IDC4DATA/SSIS_TXD/UART3_AHB_TX/GPIO13 | P19 | ISP_MP_I2C.DAT | DAT | ISP_MP_I2C |
| SSI3MISO/IDC5CLK/SSIS_RXD/UART3_AHB_CTSN/GPIO14 | N19 | DES_ISP_MFP8 | | DES_ISP_MFP8 |
| SSI3EN0/IDC5DATA/SSIS_EN/UART3_AHB_RTSN/GPIO15 | N18 | DES_EN | | DES_EN |
| | | | TP1232 TP1235 | power domain 1V8 |
| CAN0_TX/GPIO99 | E9 | PHY_RST | | PHY_RST |
| CAN0_RX/GPIO98 | D10 | ISP_PHY_GPIO9 | | ISP_PHY_GPIO9 |
| CAN1_TX/GPIO101 | E10 | PHY_INT_N | | PHY_INT_N |
| CAN1_RX/GPIO100 | E8 | | | power domain 1V8 |
| PWM0/VIN_STRIG0/GPIO18 | W16 | POC_bit30 | | |
| PWM1/VIN_STRIG1/GPIO19 | Y19 | POC_bit31 | | |
| PWM2/WDT_EXT_RST_L/GPIO20 | V14 | ISP_WDI_PMIC | R1214 1k 63mW 1% | WDI_PMIC |
| TIMER0/TM11_CLK/WDT_EXT_RST_L/PWM3/SDIO0_HS_SEL/GPIO0 | W12 | | | |
| CEHU_SAFETY_ERR_L_0 | B10 | CEHU_0 | TP1228 | |
| CEHU_SAFETY_ERR_L_1 | P17 | CEHU_1 | TP1229 | |
| TEST_MODE | T17 | ISP_TEST.MODE | R1218 10k 63mW 1% | |
| USB.D_N | AA10 | | | |
| USB.D_P | Y10 | | | |
| USB.DETECT | C10 | | | |
| USB_REXT | W10 | | | |
| ISP_JTAG.RST_L | E6 | | | |
| ISP_JTAG.CLK | J5 | | | |
| ISP_JTAG.TMS | E7 | | | |
| ISP_JTAG.TDI | G5 | | | |
| ISP_JTAG.TDO | H5 | | | |
| UART.RX | W13 | | | |
| UART.TX POC_bit15 | U14 | | | |
| UART0.RX | AA18 | | | |
| UART0.TX POC_bit13 | Y16 | | | |
| ISP_HW_ID_3 | U13 | | | |
| ISP_HW_ID_4 | V13 | | | |
| POC_bit14 | V12 | | | |
| ISP_HW_ID_1 | V17 | | | |
| ISP_HW_ID_2 | W14 | | | |
| XI_RTC | K4 | | | |
| | J4 | | | |
| XIN | A9 | | | |
| | B9 | | | |
| ISP_RST | Y20 | | | |
| RGMII.TXC POC_bit21 | A13 | | | |
| RGMII.TX_CTL | K19 | | | |
| RGMII.TXD0 POC_bit0 | G17 | | | |
| RGMII.TXD1 POC_bit1 | H18 | | | |
| RGMII.TXD2 POC_bit2 | F17 | | | |
| RGMII.TXD3 POC_bit3 | E17 | | | |
| RGMII.RXD0 | L19 | | | |
| RGMII.RXD1 | H17 | | | |
| RGMII.RXD2 | K17 | | | |
| RGMII.RXD3 | J17 | | | |
| RGMII.RX_CTL | K18 | | | |
| RGMII.MDC | L18 | | | |
| RGMII.MDIO | J18 | | | |
| RGMII.RXC | B11 | | | |
| RGMII_EXT_OSC | A12 | | | |
| ISP_PHY_GPIO1 | L17 | | | |
| POC_bit10 / USB.BOOT | B12 | | | |
| | E15 | | | |
| POC_bit4 | E14 | | | |
| POC_bit5 | D14 | | | |
| POC_bit6 | E16 | | | |
| | D13 | | | |
| DES_ISP.MFP7 | C11 | | | |
| | E13 | | | |
| | E12 | | | |
| ISP_SHUT.CTL | C14 | | | |
| | A10 | | | |
| PWR_CAM_EN_N | C12 | | | |
| PWR_SHUT_EN_N | C13 | | | |
| | A11 | | | |
| ENET1_PTP_PPS | D12 | | | |

## HW ID

## RGMII EXT CLK

## Microphone+Converter

**Mic-I²C Address: 0x35**

## IMU

**IMU- I²C Address: 0x4E**

## I²C

## RTC

## CLK

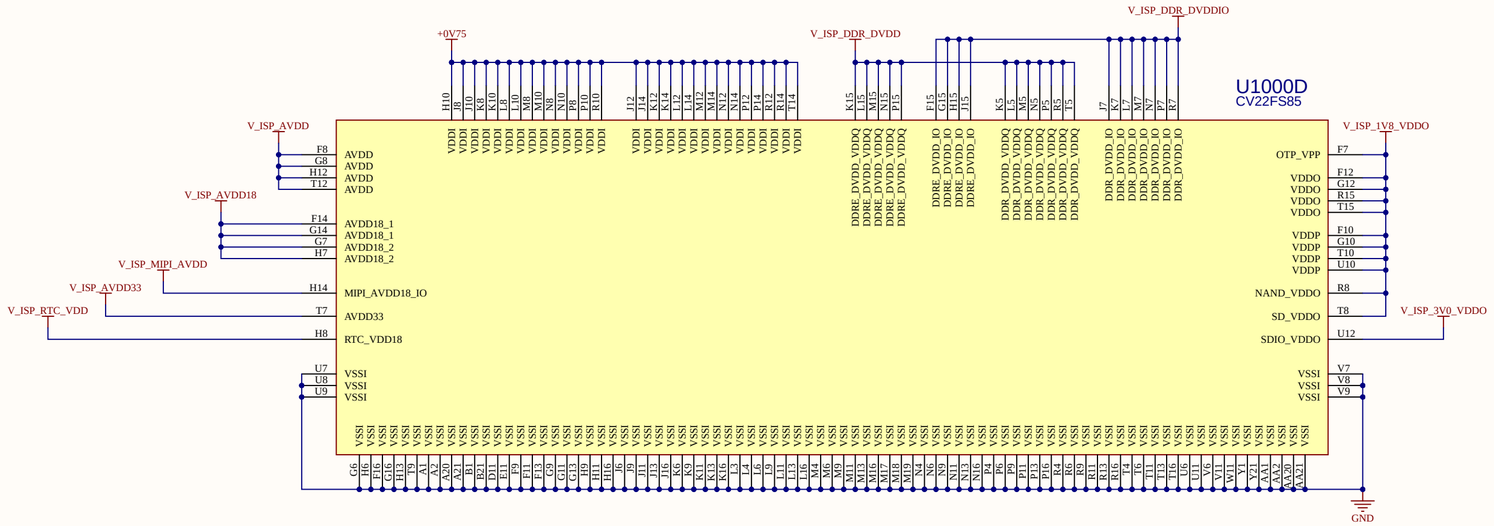
## SPI_NOR Boot:

| 0 | 1 | 2 | 3 | | 4 | 5 | 6 | 7 | | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | | 21 | 30 | 31 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 1 | | 0 | 0 | 0 | X | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | | 0 | 0 | 0 |

| | |
|---|---|
| MEKRA LANG Germany | |
| MEKRA Lang GmbH & Co. KG D - 91465 Ergersheim | |
| Bestückungsvariante / Option | 02[100Base-T1 STD] |
| Projektnummer / Project number | 11-E-01405 |
| Benennung / Designation | MVCU01.PrjPcb SOW2 - Rear and Trailer Camera |
| Revision | K |
| Blatt / sheet | 12 of 13 |
| | ISP_General.SchDoc |

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A2







| | | | |
|---|---|---|---|
| MEKRA Lang GmbH & Co. KG D - 91465 Ergersheim | Bestückungsvariante / Option: 02[100Base-T1 STD] | | |
| | Projektnummer / Project number: 11-E-01405 | Revision: K | Blatt / sheet: 13 of 13 |
| | Benennung / Designation: MVCU01.PrjPcb SOW2 - Rear and Trailer Camera | | ISP_Power.SchDoc |

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